CERAMIC TECHNICAL DATA SHEET

KTIAL

Aluminum Titanate

KEY PROPERTIES

- High thermal shock resistance
- Very low thermal conductivity
- Low coefficient of thermal expansion
- Corrosion and wear resistance
- Non-wetted by non-ferrous molten metals

APPLICATIONS

- Foundry technology
- Riser tubes
- Sprue nozzles

MAIN PROPERTIES

Property	Value	Unit
Density	3.1	g/cm^3
Flexural strength	30	MPa
Hardness	14	GPa
Young Modulus	20	GPa
Thermal Conductivity	2	W/m·K
CTE (20–1000ºC)	1.5	$x10^{-6}$ K^{-1}

** All properties measured at 20ºC unless othervise stated.*
*** Measured in a wall thickness of 2 mm*

OPERATING CONDITIONS

Max working temp (air)	1400	ºC
Recomended temp	<1100	ºC
Continuous temp	1100	ºC
Max Peak temp	1400	ºC

ELECTRICAL BEHAVIOR

Insulator

MATERIAL INFORMATION

Material type	Aluminum Titanate
Structure	Low expansion ceramic
key feature	Thermal shock resistance
Machinability	Low
Color	Light grey

Local overheating may occur at electrical contact points due to current concentration and geometry effects.

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